Polylactic-*Co*-Glycolic Acid (PLGA)

Konstantinos Avgoustakis
Laboratory of Pharmaceutical Technology, Department of Pharmacy, University of Pat
Rio, Greece

INTRODUCTION

Poly(lactide-*co*-glycolide) (PLGA) is a synthetic copolymer of lactic acid (α-hydroxy propanoic acid) and glycolic acid (hydroxy acetic acid). Lactic acid contains an asymmetric carbon atom, and therefore has two optical isomers: L(+) lactic acid and D(−) lactic acid. Lactic acid is present in nature as either an intermediate or an end product in carbohydrate metabolism. It is widely distributed in all living creatures (man, animals, plants, and microorganisms). Glycolic acid occurs in nature to a limited extent. Poly(lactide-*co*-glycolide) degrades in vivo to innocuous products. Its final degradation products are lactate (salt form of lactic acid) and glycolate (salt form of glycolic acid).

Poly(lactide-*co*-glycolide) can be synthesized by direct polycondensation of lactic acid and glycolic acid. However, the most efficient route to obtain high-molecular weight (MW) copolymers is the ring opening polymerization of lactide and glycolide. Lactide and glycolide are the cyclic diesters of lactic acid and glycolic acid, respectively, and they are prepared by thermal degradation of lactic and glycolic acid oligomers, respectively. Being a biocompatible and biodegradable polymer with adjustable properties and capable of being processed to form a variety of objects, PLGA finds extensive biomedical applications, such as sutures, orthopedic fixation devices, and drug delivery systems. Poly(lactide-*co*-glycolide) scaffolds are investigated for tissue engineering applications. The synthesis, properties, and biomedical applications of PLGA are discussed in this entry.

SYNTHESIS OF PLGA

Poly(lactide-*co*-glycolide) can be prepared by the direct polycondensation of lactic and glycolic acid (Scheme 1). The polycondensation reaction can be effected in

quired for the
icult to achieve,

acid) by azeotroj
L-lactic acid and
20–40 hr at 130°C
The solvents intr
control and purifi
fore, polycondens
the melt state. T
involves two equili
ester formation and
diester formation.
the cyclic diesters
are formed by dep
conditions of high
induce not only de
cyclic diesters in
prevents further g
high-MW polymer

To overcome tl
densation method
In the first step, oli
of monomers. In
are melt-polymeri
for example, zinc
dihydrate/*p*-toluen
step then follows i
at the solid stat
(0.5 Torr). It is beli
merization reactio
tion or other sid
polycondensation
parable to the poly
polymerization of

To prepare hig
reaction time, it is
ing polymerizatior
glycolide (Scheme
duced by thermal
acid oligomers, r
available. As the
metric carbon aton
diastereoisomers (
mate (50/50 mixt
referred to as DL-la